

Clinical	Symptom	What's wrong	Lecture
Esophageal stenosis		Partial recanalization of esophagus or deviation of trachea-esophageal septum; narrowing of lumen	1
Esophageal atresia		Complete absence of lumen occurs due to total failure of recanalization	1
Trachea-esophageal fistula	Regurgitation & choking when swallowing, polyhydramnios, excess fluid in mouth leading to pneumonia	Abnormal communication b/t trachea & esophagus; defected trachea-esophageal septum; 30% related with VACTERL	1
Congenital hypertrophic pyloric stenosis	Distended stomach, projective vomit not bile stained	Hypertrophy of muscles in pyloric region causes stenosis of lumen of pyloric canal	1
Duodenal stenosis	Vomiting bile stained	Partial occlusion of lumen due to incomplete recanalization	1
Duodenal atresia	Vomiting immediately after birth	Complete occlusion of lumen due to failure to recanalize	1
Annular pancreas		Ring of pancreas encircles 2 nd part of duodenum & causes obstruction; most likely due to bifid ventral pancreatic bud	1
Nonrotation of midgut	asymptomatic	Small intestine lies to R (normally lies to L while large intestine is to the R)	1
Reversed rotation of midgut		Midgut rotates CW, duodenum lies anterior to superior mesenteric a. & transverse colon lies posterior to SMA	1
Subhepatic cecum & appendix	Difficulty in diagnosis of appendicitis	Cecum gets adherent to liver & doesn't descend to iliac fossa	1
Mixed rotation & volvulus of midgut		Cecum lies inferior to pylorus & is fixed to posterior abdominal wall by peritoneal bands, causing duodenal obstruction	1

Omphalocele		Persistence of abdominal herniation; loops of midgut did not return to abdominal cavity so organs are outside of body in thin sac	1
Umbilical hernia		Protrusion of skin & subcutaneous tissue no intestines/organs	1
Gastroschisis	worst out of 3 (previous 2) intestines grow outside body in amniotic fluid	Defect near median plane of abdominal cavity = LATERAL FOLDS FAIL TO CLOSE	1
Meckel's diverticulum	Mimics signs of appendicitis & peptic ulcer & can produce periumbilical pain & internal bleeding; if presents as indirect inguinal hernia, called "hernia of Littre"	Fingerlike projection from ileum, due to retention of embryonic intestinal stalk; secretes gastric & pancreatic juices	1
Vitelline cyst		When Meckel's diverticulum attaches to umbilicus and forms a cyst	1
Umbilical (umbilicoileal) fistula		Meckel's diverticulum attaches to umbilicus and remains patent creating external opening at umbilicus	1
Megacolon	Severe colonic dilation proximal to affected area; buildup of fecal material	Absence of enteric ganglia due to failure of neural crest cells to form myenteric plexus causing a nonfunctional segment of bowel; can also be caused by parasitic disease	1,5
Pernicious anemia		Lack of intrinsic factor (which binds to vitamin B12)	3
Achalasia		Failure of lower esophageal sphincter to relax leading to impaired peristalsis & accumulated food in esophagus; usually defective ENS	6
Sigmoid volvulus		Twisting of sigmoid colon around mesentery, causing obstruction of colon & intestinal ischemia	8
Pyloric stenosis	Projective vomiting; dehydrated, always hungry	Hypertrophy of pyloric sphincter, occluding sphincter	11

Dumping syndrome	Rapid gastric emptying, distension of proximal SI, diarrhea & abdominal discomfort	Develops after surgery, food moves from stomach to SI too quickly; Autonomic reflexes >> GIT hormones	11
Gastric & duodenal ulcers	Can coexist together	NSAIDs can induce gastric ulcers bc block PG synthesis; duodenal ulcers are distal to pyloric sphincter; chronic ulcers extend to muscularis	11
Peptic ulcer disease	Recurrent & episodic epigastric pain when food intake; anemia	chronic	11
Cystic Fibrosis	Decreased Na ⁺ & H ₂ O movement, decreased pancreatic duct secretion; chronic pancreatitis, steatorrhea, undernourished	Defective CFTR channel protein; creating large accumulation of Cl ⁻ in the pancreatic ductal cell	12
Jaundice	Yellowish discoloration of skin, mucous membranes, sclera, & nail beds	Binding of bilirubin to CT ; bilirubin levels are >2 mg/dL (normal is <1)	13
Prehepatic jaundice	Increase total bilirubin, increases unconjugated bilirubin, no urine bilirubin , and increase urobilinogen	Increased hemolysis of RBC leading to increase unconjugated bilirubin	13
Hepatocellular jaundice	Conjugated bilirubin in urine because it is flowing over into blood = orange urine	Liver cell does not fxn properly so decreased uptake of unconjugated & decreased secretion of conjugated	13
Posthepatic jaundice	obstruction of bile duct so no bile enters intestine so decrease urobilinogen and stercobilin so feces is colorless	Decrease excretion of bilirubin via bile; urine bilirubin is present but urine urobilinogen is absent	13
Kernicterus	Lethargy, altered muscle tone, permanent neurological issues	Jaundice in newborns = increase unconjugated bilirubin bc now activity of hepatic UDP-glucuronyl transferase; unconjugated deposits in brain	13
Crigler-Najjar syndrome type I	Inherited unconjugated hyperbilirubinemia; usually fatal if not treated	Most severe; complete deficiency of UDP glucuronyl transferase; kernicterus & developmental delay	13

Crigler-Najjar syndrome type II	Inherited unconjugated hyperbilirubinemia; jaundice	Low activity of bilirubin glucuronyl transferase; children respond to phenobarbital	13
Gilbert's syndrome	Inherited unconjugated hyperbilirubinemia; mild jaundice usually following infection or stress	Bilirubin UDP-glucuronyl transferase activity is 50%	13
Dubin-Johnson Syndrome	Inherited conjugated hyperbilirubinemia; young adults	Conjugated bilirubin is in the blood because ABC transporter is not working properly to get bilirubin into bile	13
Gallstones	"fair, fat, 40 YO female" lots of pain!! (episodic)	Excess cholesterol aggregates to form crystals	14
Biliary colic		Caused by spasm of smooth muscle of gallbladder in attempt to expel gallstones	20
Acute cholecystitis	Pain in R upper quadrant which can cause subdiaphragmatic parietal peritoneum irritation -> pain in R shoulder (referred due to phrenic n)	Due to gallstones	20
Hemochromatosis	Toxic levels of Fe in liver & tissue damage if stored in tissues; monitor plasma ferritin levels	Excessive absorption of Fe in GIT	15
Cystinuria	Formation of kidney stones	Defect in basic AA uptake	15
Pernicious anemia	Vit B12 deficiency = tiredness & fatigue	Deficiency of vit B12 due to lack of intrinsic factor from gastric atrophy or gastrectomy	15
Cirrhosis of liver	Liver tissue becomes ischemic & we get portal hypertension	Hardening of liver (replaced by fibrotic tissue) causing progressive destruction of hepatocytes which impedes blood circulation	15
Biliary colic		Caused by spasm of smooth muscle of GB in attempt to expel gall stonefs	16
Acute cholecystitis	Pain in R upper quadrant (referred pain to R shoulder)	Once biliary colic has pushed gall stone into cystic duct	16

Splenomegaly	Portal hypertension	Enlargement of spleen due to diseases affecting reticulo-endothelial system	16
Spinal cord injury	Loss of voluntary control of defecation causing automatic defecation when rectum fills with feces	Descending common pathway controlling external sphincter (corticospinal tract) severed	17
Lactose intolerance	Cramps, diarrhea, & flatulence	Deficiency of not properly digesting lactose which results in lactose passing on to LI; bacteria in LI (osmotically active) degrade lactose -> large vol. H2 gas & CO2	18
Primary lactose intolerance	Start to show lactose intolerance symptoms around 7 or 8; diet should be low in lactose or can take lactase pills prior to ingestion	Lactase declines with age (age-dependent)	18
Secondary Lactose intolerance	Lactose intolerance symptoms due to loss of enzyme	Occur due to loss or damage of intestinal mucosal cells (food poisoning, severe diarrhea, gastroenteritis, celiac disease)	18
Cholelithiasis	Pain in R quadrant	Cholesterol gallstone disease due to reduction of hepatic bile salt secretion OR increased biliary cholesterol secretion leading to less soluble cholesterol (bc not enough bile salts to make micelle)	19
Pancreatitis	Severe inflammation	Gallstones block pancreatic duct preventing the enzymes to be released into duodenum; trypsin is active within pancreas, activating all other proteases; they're not secreted into duodenum, thus decreasing digestion of all food groups	19, 22
Barret's esophagus		Stratified squamous to simple columnar at the squamo-columnar jxn	20
Peptic ulcers	If chronic, can erode artery causing bleeding into stomach -> vomit blood; can cause loss	- Occur most commonly in first part of duodenum due to flow of acid through pyloric valve	20, 22

	of parietal cells -> loss intrinsic factor -> pernicious anemia	- Also occurs on lesser curvature of stomach - Anterior duodenal ulcers are on peritoneal cavity = peritonitis	
Chron's Disease	Ulcers with normal mucosa b/t; fibrosis due to chronic inflammation results in strictures -> obstruction	Ulcer formation of mainly terminal ileum but also affects LI & upper GI	22
Ulcerative colitis	Flare ups create shallow crypt abscesses due to neutrophils	Chronic bouts of inflammation & ulcer formation followed by healing	22
Leigh's syndrome	Leads to lactic acidosis	Pyruvate dehydrogenase deficiency	28
Thiamine deficiency	Leads to lactic acidosis	Lack in B1	28
Von Gierke	Leads to lactic acidosis	Defect in lactate -> pyruvate	28
Glucokinase deficiency in beta cells of pancreas (MODY-2)	Impaired insulin secretion from beta cells; chronic mild hyperglycemia bc insulin is not secreted until glucose levels >7-8 (normal >5) ; NOT overweight ; type 1 diabetes symptoms	Genetic deficiency of pancreatic glucokinase can lead to maturity onset diabetes of the young type 2 (MODY-2)	30
Wernicke-Korsakoff	Ataxia, ophthalmoplegia, memory loss, cerebral hemorrhage, heart failure, decrease ATP, increase cardiac output (wet beri-beri)	Thiamine deficiency -> low PDH activity; at risk: alcoholics & malnourished	31, 54
PDH deficiency	Lactic acidosis, neurological defects, myopathy, flared nares, wide nasal bridge, corpus callosum agenesis, cerebral & basal ganglia cysts; normally fatal at early age	Increase pyruvate -> increase in lactic acid & alanine (via transamination); decrease acetyl CoA, decrease ATP	31
Heavy metal poisoning (arsenate, mercury, lead)	CNS majorly effected bc it cannot use fatty acid as fuel source; treatment: give lipoic acid	Bind to -SH group (on lipoic acid); PDH complex becomes inactive	31
Glycogen synthase deficiency (rare)	Glycogen storage disease (type 0)	Missing glycogen synthase - cannot make glycogen	34

Von Gierke disease	Glycogen storage disease (type 1); severe hypoglycemia bc glycogen degradation & gluconeogenesis are impaired; lactic acidemia (need to make free glucose somehow); hyperuricemi; progressive renal disease; lactic acidosis Type 1a: deficiency of G6P in ER membrane Type 1b: deficincey of G6P translocase into ER	<i>Glucose 6 phosphatase deficiency;</i> hepatonephromegaly with high glycogen content in the liver & kidney; treatment = nocturnal gastric glucose infusions	34
Pompe disease	Glycogen storage disease (type 2); lysosomal glycogen accumulation in heart, muscle, kidney & liver; mainly affects muscle tissue; infant – early death due to heart failure; normal blood glucose levels & normal glycogen cytosolic amounts & cardiomegaly & heart failure	<i>Lysosomal alpha glucosidase (aka lysosomal acid maltase aka 1-4 glucosidase) deficiency;</i> defective glycogen degradation in lysosomes by acid maltase (aka alpha1-4 glucosidase = cleaves glycogen into free glucose), so it accumulates within the lysosomes; treatment = enzyme replacement therapy by infusion	34
Cori disease (aka Forbes disease or limit dextrinosis)	Glycogen storage disease (type 3); muscular weakness, hypotonia, mild hypoglycemia (can break down some, just not past limit dextrin) & muscular dystrophy (weakness); hepatomegaly; patients may need wheelchair later in life; considered muscular dystrophy	Debranching enzyme deficiency (4:4 transferase activity) – have limit dextrin; glycogen has abnormal structure	34
Andersen disease	Glycogen storage disease (type 4); infantile hypotonia, infantile liver cirrhosis (due to insoluble glycogen) & hepatomegaly, scarring of liver, death by 5 years of age	Branching enzyme deficiency (4:6 transferase); abnormal glycogen structure with long glucose chains with less branches; abnormal structure	34

		causes attack of autoimmune system because the glycogen is abnormal.	
Mc Ardle syndrome	Glycogen storage disease (type 5); child appears tired & unmotivated so hard to diagnose; Myoglobinemia, Myoglobinuria (breakdown); rhabdomyolysis after forced exercise due to lack of ATP – also lactate is NOT increased; CK-MM increased; high glycogen levels with normal structure; normal blood glucose with muscle weakness	Myophosphorylase deficiency; abnormal glycogen degradation in muscle ; liver isozyme is normal	34
Hers disease	Glycogen storage disease (type 6); hepatomegaly, growth retardation, mild fasting hypoglycemia (bc gluconeogenesis is still functioning), high glycogen with normal structure	Hepatophosphorylase deficiency; abnormal glycogen degradation in the liver	34
Tarui disease (rare)	Glycogen storage disease (type 7); clinically similar to McArdle leading to muscle cramps; mild hemolysis occurs in this disease due to 50% PFK1 deficiency in RBC	PFK-1 deficiency in RBC (ML type) & muscle (MM type); liver isozyme is normal	34
Benign fructosuria (essential fructosuria)	Asymptomatic because no toxic metabolites accumulate in liver	Deficiency of fructokinase in liver; when ingest sucrose, fructose cannot be metabolized, so it is found in urine	36
Hereditary fructose intolerance	Gluconeogenesis & glycogenolysis inhibited bc low ATP, leading to hypoglycemia in fed state ; PTs who continue to eat this will acquire hepatocellular failure -> jaundice ; <i>Hypoglycemia & liver disease</i>	Deficiency of aldose B in liver; ingestion of sucrose, fructose, & sorbitol lead to trapping of fructose 1 phosphate in hepatocytes -> ATP deficiency	36
Nonclassical galactosemia	accumulation of galactitol results in <i>cataract</i> formation in children; symptoms less severe than classical form;	Deficiency of galactokinase;	36

Classic galactosemia	Gluconeogenesis & glycogenolysis inhibited bc low ATP, leading to hypoglycemia in fed state ; galactose 1 phosphate and galactitol accumulation leads to jaundice & hepatomegaly ; <i>cataracts, liver disease, & developmental delay</i>	Deficiency of galactose 1-phosphate uridyl transferase (GALT);	36
Cushing syndrome	Hyperglycemia = secondary diabetes	Excess cortisol; cortisol induces enzyme synthesis of gluconeogenesis, increasing blood glucose	37
Addison's disease	Hypoglycemia	Low cortisol	37
Hypoglycemia	Dizziness, tachycardia, seizures	Medical emergency bc brain needs glucose, cannot run off ketone bodies	37
Acute alcohol intoxication	Hypoglycemia		
Leber's Hereditary Optic Neuropathy	25 – 35; leads to legal blindness; normally affects Complex I	Defect in NADH dehydrogenase (ETC); degeneration of retinal ganglion cell & atrophy of optic nerve	38
Kearns-Sayre syndrome	Onset before 20YO; paralysis of eye muscles & degeneration of retina; mental retardation; muscle weakness	Deletion in mtDNA	38
Deafness induced by aminoglycoside antibiotics	Hearing loss & deafness induced days after aminoglycoside antibiotic administration	Mitochondria rRNA that is inherited mutation that causes predisposition to aminoglycoside ototoxicity	38
MELAS syndrome	Mitochondria myopathy, encephalopathy, lactic acidosis, stroke like episodes	Most common mitochondrial disease; oxidative phosphorylation defect; mutation in mt tRNA; excess pyruvate to be reduced to lactic acid with accumulates in blood & other fluids	38
MERRF syndrome	Myoclonic epilepsy, ragged red fibers; onset 6-16 YO; gomoritrichrome stain	Most common mutation in mt tRNA gene	38

Smith-Lemli-Opitz Syndrome (AR)	double bond formation in one of the rings does not occur; decreased cholesterol; high infant mortality rate; heart defects, malformed limbs, growth retardation, microcephaly; cholesterol administration helps with growth but not CNS defects due to embryonic microcephaly	Deficiency of 7 dehydrocholesterol reductase which is responsible for final step in production of cholesterol; loss of function mutations in DHCR7 gene	39
Cholesterol gallstone disease (cholelithiasis)	Bile becomes solid within the gallbladder; treatment: removal of gallbladder or administration of chenodeoxycholic acid	Precipitation of cholesterol (in bile) in the gall bladder due to deficiency of lecithin and/or bile salts	39
MCAD deficiency (AR)	Systemic FA oxidation disorder; 6-24 months of age; severe hypoglycemia in response to fasting or illness bc tissues rely on glucose as source of energy since beta oxidation cannot occur	Decreased ability to oxidize fatty acids with 6-10C chains; medium chain acyl CoA excreted in urine; dicarboxylic acids also found in urine bc increased omega oxidation	41
Carnitine deficiency	Systemic FA oxidation disorder; Presents at early age; hypoglycemia & ketogenesis decreased → hypoketonemia; increase lipid TAGs all over; CNS affected (unconscious) Myopathic disorder; Later onset; muscle weakness & cardiomyopathy; CK-MM present & myoglobin in urine; increase lipid TAGs in muscles	Carnitine uptake into tissues is impaired therefore transport of long chain FA into mit. Is impaired & beta oxidation decreased	41
CPT-I deficiency	Systemic FA oxidation disorder; Hypoglycemia & hypoketosis; increased carnitine levels	Affects liver	41

CPT-II deficiency	Mypoathic FA oxidation disorder; Cardiomyopathy & muscle weakness; lipid deposits found in skeletal msc ; prolonged exercise results in myoglobinuria & increased CK-MM	Affects muscle	41
Jamaican vomiting sickness	Ingestion of unripe ackee fruit results in hypoglycemia & vomiting; can lead to coma & death if not immediately treated	Hypoglycin A is an inhibitor of MCAD enzyme, inhibiting beta oxidation	42
Zellweger syndrome	Delayed development, hepatomegaly, hepatocellular failure, fatal in infancy; all peroxisomal enzymes lost (vs. Refsum where only alpha oxidation is lost)	Normally: Very long chain FA are initially oxidized in the peroxisomes & the shorter FA are transported to mitochondria; this syndrome has defective peroxisomal biogenesis mainly affecting liver & brain, 26C FA circulating in blood causing neurological damage & demyelination	42
Refsum disease	Visual defects, ataxia, & polyneuropathy & skeletal manifestations; management: dietary restriction of branched chain FA (no dairy)	Deficiency of peroxisomal phytanyl CoA alpha hydroxylase (defect in alpha oxidation); Phytanate accumulates in tissues (especially neurologic)	42
Cystinuria	Cystine excreted in urine & can precipitate in renal tubules forming stones; hexagonal cystine crystals in urine do not confuse with sodium urate of PTs with gout	Tubular reabsorption of cystine decreased due to inherited deficiency of cystine transporter	44
Hartnup disease	Decreased dietary absorption of tryptophan and increased excretion in urine; both this & cystinuria have messed up reabsorption of filtered AA from kidney	Inherited defect in transport of tryptophan cannot uptake AA; can lead to NAD+ deficiency (pellagra)	44

Niacin deficiency	Pellagra (rough skin) 4D's: diarrhea, dermatitis, dementia, death	Decrease of niacin (because it comes from tryptophan & diet) causing NAD+ deficiency	44, 54
Hyperammonemia I	Leading to elevated ammonia; sometimes responds to arginine intervention because it can stimulate formation of NAG which might stimulate deficient CPS1	Deficiency of carbamoyl phosphate synthetase 1 in urea cycle	45
Hyperammonemia II (X linked)	Leading to orotic acid excreted in urine & elevated serum ammonia; most common	Deficiency of ornithine transcarbamylase in urea cycle	45
Citrullinemia	Citrulline excreted in urine; hyperammonemia	Deficiency in argininosuccinate synthetase in urea cycle	45
Argininosuccinic aciduria	Argininosuccinate excreted in urine	Deficiency in argininosuccinate lyase in urea cycle	45
Argininemia	Arginine levels increase the only time you do not try to treat with arginine bc you already have increased arginine levels	Deficiency in arginase in urea cycle	45
Acquired hyperammonemia	Ammonia produced in intestine directly enters circulation & results in neurotoxicity (NH3 intoxication)	Liver disease due to viral or drug induced hepatitis or alcoholic cirrhosis	45
Hurler syndrome (AR)	<i>Mucopolysaccharidoses</i> ; accumulation of heparan sulfate & dermatan sulfate; corneal clouding , mental retardation, dwarfing, upper airway obstruction, hearing loss, hepatosplenomegaly	Alpha-L-iduronidase deficiency; urine positive for GAGs; enzyme replacement therapy with iduronidase	46
Hunter syndrome (X linked)	<i>Mucopolysaccharidoses</i> ; accumulation of heparan sulfate & dermatan sulfate; physical deformity & mental retardation, hepatosplenomegaly; Mongolian blue spot on back when born	Iduronate sulfatase deficiency	46
Tay sach's disease (AR)	<i>Sphingolipidoses</i> ; progressive neurodegeneration after 3-6 months causing	Beta hexosaminidase A deficiency; accumulates ganglioside (GM2)	46

	blindness; born looking normal then developmental milestone delay often regression; fatal by 6; cherry red spot on macula		
Gauchers disease (AR)	<i>Sphingolipidoses</i> ; macrophages engorged with glucocerebrosides; adult form shows no neurological damage but marked hepatosplenomegaly & osteoporosis of long bones – “bone crisis”	Beta glucosidase deficiency; most common lysosomal storage disorder; accumulates glucosyl ceramide (aka glucocerebroside)	46
Fabry disease (X linked)	<i>Sphingolipidoses</i> ; peripheral neuropathy & GI tract issues; globoside accumulates in BV causing skin rash; farting issue ; pain in extremities, red/purple skin rash	Alpha galactosidase deficiency; accumulation of globoside (aka ceramide trihexoside)	46
Niemann-Pick disease (AR)	<i>Sphingolipidoses</i> ; type A is severe infantile form (cherry red spot in macula...like tay sachs); type B appears later in childhood with hepatosplenomegaly (...like gaucher)	Sphingomyelinase deficiency; accumulation of sphingomyelin in neuronal tissues	46
Metachromatic leukodystrophy	<i>Sphingolipidoses</i> ; born normal but progressive paralysis & demyelination	Aryl sulfatase A deficiency; accumulation of sulfatide (rich in neurons)	46
I cell disease	Deficiency of many lysosomal enzymes & accumulation of GAGS & sphingolipids in lysosomes causing inclusions in fibroblasts; large hepatosphenomegaly; limitations of hips & knees	Defect in trafficking enzymes to lysosomes bc golgi fails to add mannose 6 phosphate marker to enzymes so they accumulate in blood stream	46
Phenylketonuria (PKU) (AR)	Elevated PHE, mental issues, seizures, autistic behaviors, hypopigmentation (bc no melanin) ; avoid eggs, milk, meat, & aspartame; major IQ reduction;	Deficiency in phenylalanine hydroxylase (PAH) ; tyrosine not made (can give as essential aa) so decreased melanin synthesis; phenylpyruvic acid accumulation in urine (mousey odor)	47

	Treatment: avoidance & sapropterin which is cofactor of PAH & high doses can cause conversion of PHE -> TYR	Maternal PKU: moms must maintain low PHE levels during pregnancy bc can cause microcephaly & neurological issues (teratogenic) for fetus even though they are normally heterozygous	
Alkaptonuria	1 st inherited enzyme deficiency found in humans; causes homogentisic acid deposits in cartilage & CT tissues (ochronosis) causing dark discoloration of cartilage; also excreted in urine causing brown discoloration; can lead to arthritis later in life	Deficiency in homogentisic acid oxidase ; cannot eat PHE or TYR bc this is downstream of TYR degradation	47
Tyrosinemia Type I (Tyrosinosis)	Liver & renal failure; cabbage odor of urine; usually fatal; cannot eat PHE or TYR; cannot form ketone bodies	Deficiency in fumaryl acetoacetate hydrolase ; causing build up of fumaryl-acetoacetate causing cabbage like smell to urine	47
Maple syrup urine disease	Urine smells like maple syrup; infants; poor feeding, vomiting, poor weight gain; neurological signs; branched chain AA also buildup causing neurological manifestations Treatment: diet restriction of branched chain AA (leucine, isoleucine, valine) which is hard bc catabolic states create these from protein sources when fasting; supplementation of TPP may be useful to activate enzyme	Deficiency in branched chain alpha keto acid dehydrogenase (requires TPP) ; leading to buildup of branched chain keto acids which cause ketosis & maple syrup odor of urine	47
Methylmalonic Aciduria	Feeding problems, refusing to suck & then vomiting of infant; breathing problems; v dehydrated; metabolic acidosis; neurological manifestations – seizures, encephalopathy;	Deficiency in methylmalonic CoA mutase leading to accumulation of methylmalonic acid	47

	Treatment: can improve with vit B12 (cobalamin) supplementation		
Homocystinuria	Dislocation (detached) of lens, skeletal abnormalities, mental development; arteries narrow at v young age (child); osteoporosis in children Treatment: can improve with oral vit B6 (PLP)	Group of disorders with defect in homocysteine metabolism; deficiency in cystathionine beta synthase (transulfuration pathway) ; homocysteine binds to CT & disrupts its structure; <i>any pathway using SAM as cofactor will produce homocysteine (ie creatine synthesis)</i>	47
Parkinson disease	Slowness, rigidity, tremors, cramping, involuntary head movements	Decrease dopamine in the brain Treatment: can improve with L-DOPA	48
Pheochromocytoma	Headache, palpitations, anxiety, panic attack, perspiration, HTN, tachycardia	Adrenal gland tumor upregulating catecholamines; 24 hr urine test during episodic symptom for VMA	48
Carcinoid syndrome	Cutaneous flushing, diarrhea, bronchospasm, endocardial fibrosis	Tumor of serotonin producing cells in GIT (APUD cells) bc they are taking up too much tryptophan so creating too much serotonin; increased 5HIAA in urine	48
PKU II (malignant PKU)	Very severe CNS symptoms; decreased neurotransmitter synthesis & catecholamines; hyperphenylalanemia Treatment: dietary PHE restriction & providing dietary bipterin & precursors of NT but this is difficult bc does not cross BBB	Deficiency of dihydrobiopterin synthesis or dihydrobiopterin reductase (BH2/BH4) ; BH2 is reduced by NADPH to BH4; amount of BH4 is decreased	48
Albinism	Lower visual acuity & photophobia; increased risk of skin damage due to sunlight	Deficiency of tyrosinase ; Partial or complete deficiency of melanin in skin, hair, & eyes	48
Porphyrias	Chronic or acute; neurologic dysfunction, mental disturbance, photosensitivity	Blood pigment Hb is abnormally metabolized; Grouped into hepatic or	49

	Treatment: IV injection of glucose & hemin (hepatic type) causing decrease of ALAS1; ALA increase will compete with GABA (which is normally an inhibitory so if it competes, nothing will be inhibited)	erythropoietic; lead to accumulation of intermediates of heme synthesis - colored intermediates give problems	
Acute intermittent porphyria (AD) Hepatic	Porphobilinogen accumulates in blood & urine; once urine contacts air/light, changes to black – oxidation of porphobilinogen; severe abdominal pain, tachycardia, respiratory problems, confusion, not photosensitive bc HMB not formed; acute onset due to drugs that stimulate CytP450 synthesis, infections, alcohol, abnormal estrogen	Deficiency of HMB synthase (porphobilinogen deaminase); ALA accumulates as heme is lacking & ALA synthase is stimulated; high ALA & porphobilinogen levels interfere with GABA; do not prescribe barbituates bc causes stimulation of CytP450 Treatment: glucose infusion bc hypoglycemia induces ALAS1 synthesis but we don't want that bc we will just cause more issues so glucose will inhibit it; if v severe: hemin IV	49
Congenital erythropoietic porphyria (AR) Erythropoietic	Chronic; PTs photosensitive because HMB is spontaneously closed to type I which leads to red colored molecules leading to extremely severe & painful photosensitivity; onset = childhood or earlier; blisters, poor wound healing, & ulcers	Deficiency in uroporphyrinogen III synthase; hereditary defect only in bone marrow; uroporphyrinogen I & coproporphyrinogen I accumulate in urine, tissue, & blood	49
Porphyria cutanea tarda Hepatic	Photosensitive; urine is red upon release in natural light but pink in fluorescent light; chronic; can be acquired due to disease such as hepatitis or alcoholism; symptoms expressed when hepatic iron overload, sunlight, or excessive alcohol Treatment: avoid those 3 things	Deficiency of uroporphyrinogen decarboxylase (type I: sporadic, type II: familial); uroporphyrin III accumulates in liver, blood, & skin; most common form	49

Hereditary deficiency of ALDH-2	Acetaldehyde in blood leads to flushing & elevated heart rate, nausea & vomiting	Deficiency of mitochondrial acetaldehyde dehydrogenase (DH2) ; common in east Asian heritage; decreased metabolism of acetaldehyde in mitochondria; increases Km & decreases Vmax	51
Folate deficiency	Macrocytic, megaloblastic anemia – larger RBC or cells with incompletely mature nuclei; during pregnancy : associated w high risk of neural tube defects (spina bifida); FIGLU in urine	Decreased folic acid, decreased pyrimidine & purine synthesis, decrease DNA synthesis & cell division, decrease RBC formation decrease HB; impaired absorption due to intestinal mucosal disease; methotrexate is antagonist	52, 54
Pernicious anemia	Macrocytic anemia & neurological features – tingling of feet & hands, inability to walk, urinary incontinence; increase methylmalonate & homocysteine levels in urine; deficiency of usable form (folate trap)	Lack of B12 (cobalamin) due to lack of intrinsic factor production from parietal cells; at risk : PTs with parietal cell destruction OR PTs with ileomucosal disease	52
Vit A deficiency	Earliest symptom: night blindness bc regeneration of rhodopsin is delayed; xerophthalmia, keratomalacia -> opacity leading to blindness, immune deficiency; when epidemic of measles kids get Vit A shots to protect from pulmonary infections & increase innate immunity	Vitamin A deficiency; malabsorption of fats can lead to this deficiency	53
Hypervitaminosis A	Headaches , dry & pruritic skin (peeling), if pregnant -> spontaneous abortion & congenital malformation in fetus	Too much Vitamin A; liver stores vit A causing enlarged liver & benign intracranial hypertension (mimics brain tumor)	53

Rickets (Vitamin D deficiency)	Bow-leg deformity, pigeon chest deformity, frontal bossing; cartilage not being mineralized into bone	Decreased Vit D in children, reducing calcium absorption decreasing serum calcium increasing PTH increasing demineralization of bone; serum ALP increased (bone disease)	53
Vitamin D resistant Rickets	High levels of calcitriol, low serum calcium & phosphate, high 1,25 hydroxyvitamin D levels bc no feedback inhibition of calcitriol formation since it is not absorbed body thinks it still has low calcium	Inherited condition where calcitriol is unable to bind to its receptor in intestinal mucosal cells	53
Osteomalacia (vitamin D deficiency)	Serum ALP increased, can be secondary to dietary deficiency, renal disease, or liver disease	Vitamin D deficiency in adult; bones demineralize and are susceptible to fracture	53
Hypervitaminosis D	Ectopic (soft tissue) mineralization; prolonged hypervitaminosis D can lead to calcification in soft tissues like kidney	Too much Vitamin D; hypercalcemia due to increased calcium absorption from diet; increased phosphate absorption	53
Vitamin K deficiency	Increased tendency to bleed: hematuria, melena (black stools), bruises, bleeding from gums	Neonates at risk bc hemorrhagic disease & sterile intestines (no synthesis of vit K); also see in PTs with fat malabsorption (CF) & prolonged broad spectrum antibiotics	54
Vitamin C deficiency (Scurvy)	Perifollicular hemorrhages, spongy bleeding gums, bleeding into joints, frequent bruising, impaired wound healing; perifollicular hemorrhages	Defects in CT due to decreased hydroxylation of collagen results in fewer number of H bonds in collagen	54
Thiamine deficiency (BeriBeri)	Dry Beri Beri: affects neurons; polyneuropathy – disruption of motor, sensory, & reflex arcs, can progress to paralysis	Common when polished rice is major diet component bc lacks thiamine; affects highly aerobic tissues; low erythrocyte transketolase activity & increase activity on addition of TPP	54

	Wet Beri Beri: cardiovascular symptoms		
Riboflavin deficiency	Cheilosis – areas of pallor, cracks & fissures at the angles of mouth; Glossitis – inflammation & atrophy of tongue	Riboflavin deficiency	54
Pyridoxine deficiency	Microcytic anemia, peripheral neuropathy, cardiovascular disease, seizures	Cannot form PLP (vitamin B6 pyridoxine deficiency)	54
Menkes Syndrome (kinky hair) (X linked)	Hair is twisty, greyish, & “kinky”; aneurysms & cerebral dysfunction due to reduced activity of lysyl oxidase	Inherited defect in absorption of copper from GI tract (diet); copper deficiency	55
Wilson Disease (AR)	Accumulation of toxic levels of copper in liver, brain (basal ganglia), & eyes leading to cirrhosis, mood changes, & Kayser Fleisher rings (copper in cornea) Onset: 30-40YO	Disorder of copper metabolism & mutation of copper transporting ATPase in the liver so cannot attach Cu to ceruloplasmin & cannot excrete Cu to bile; lab: low ceruloplasmin in blood, increased urinary Cu, low Cu in bile, & high Cu in liver	55
Iron deficiency anemia	Chronic bleeding is important risk factor; hypochromic microcytic, fatigue, pallor, weakness, dizziness, brittle nails, pica (appetite for soil)	Reduced dietary iron absorption; at risk: infants, pregnant women, regular blood donors; lab: low Hb, low hematocrit, low MCV, low MCMH, low serum ferritin, low serum iron, low % sat. of transferrin, increase TIBC	55
Hereditary Hemochromatosis (iron overload) (AR)	Onset 40YO; males affected earlier than females; excess iron damages tissues by lipid peroxidation through free radicals & DNA damage; joint pain, chronic fatigue, cirrhosis, diabetes, cardiac dysfunction, brown skin pigmentation; lab: high serum ferritin levels, high serum iron, transferrin sat. > 50%, TIBC is decreased; liver, beta cells of pancreas & skin mainly affected	Genetic defect causing excessive absorption of iron (mutation of HFE gene) most frequent mutation is C282Y allele of HFE (allelic heterogeneity); excessive accumulation of iron in organs (mainly liver & pancreas) because body thinks it is in iron deficient state and the negative feedback does not kick in	55

[illegible]